Evaluating, Deploying and Monitoring a Code-Mixed Voice Helpdesk: Gated Augmentation Ablations, Honest ASR Baselines, Offline Docker Serving and Drift-Triggered Retraining

Rajdeep Vraj*
Indian Institute of Technology Gandhinagar
Roll No. [TBD: roll]
Gandhinagar, Gujarat, India

Vipul Sunil Patil*
Indian Institute of Technology Gandhinagar
Roll No. [TBD: roll]
Gandhinagar, Gujarat, India

Satvik Sharma*
Indian Institute of Technology Gandhinagar
Roll No. [TBD: roll]
Gandhinagar, Gujarat, India

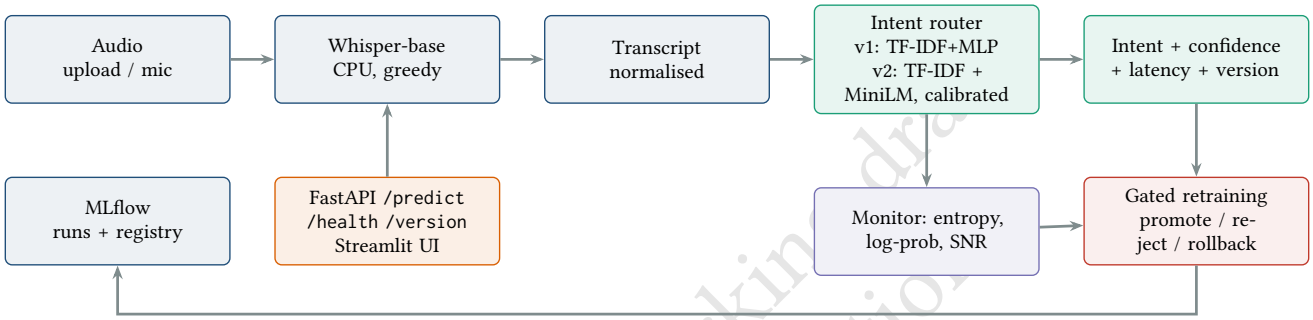


Figure 1: System architecture. Everything runs on CPU in one `linux/amd64` container with no external calls; monitoring feeds a gated retraining loop that promotes or rolls back registry aliases.

ABSTRACT

We evaluate and deploy a voice-helpdesk pipeline on two complementary tracks. **Track A** (final 182-clip consensus corpus) compares 12 data configurations for a calibrated multilingual intent classifier under 5-fold group-stratified cross-validation: the raw-transcript baseline reaches macro-F1 0.664 ± 0.071, the best augmented configuration gains only +0.013, and no configuration differs significantly (paired $p \geq 0.21$); the full "all gates active" pipeline is *worse* (−0.021). On unseen speakers (36 clips) accuracy is 75.0% (95% Wilson CI [58.9, 86.2]); off-the-shelf Whisper-base on all 182 clips gives WER 1.617 and CER 1.266, with Gujarati best (0.55) and Hindi and code-mixed speech worst (2.56, 2.08). **Track B** (deployed Whisper-base + TF-IDF/MLP router, frozen pilot gold) provides MLflow tracking over 15 runs, a FastAPI/Streamlit service with offline Docker packaging, a drift-replay monitor and a gated retraining loop; on its 29 gold clips WER is 1.272 and intent accuracy 24.0%, within chance, and proposed and baseline ASR are identical. We report effects with 95% intervals and paired tests, disclose where the evidence does not support a claim (the replay alert is schedule-driven, the retraining gate decides on one clip, container measurements are pending), and list the fixes each finding requires.

CCS CONCEPTS

• **Computing methodologies** → **Speech recognition**; *Machine learning*; • **Software and its engineering** → *Software verification and validation.*

KEYWORDS

Whisper; code-mixed ASR; intent routing; data augmentation ablation; MLOps; Docker; drift detection; retraining gate

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1 INTRODUCTION AND EVIDENCE MAP

Assignment 2 turns our data into a deployable, monitored system. Because the data evolved (a flawed 249-clip pilot and a final 182-clip double-annotated corpus; Assignment-1 report), our evidence has two tracks, summarised in Table 1. We follow the rules: choices are made on validation or cross-validation data; the frozen pilot gold set (SHA-256 `64b7647b. . . 48ca9`) is never used for tuning and every use is logged in `TEST_SET_LOG.md`; and every number is traceable through the evidence index (Appendix A).

*All authors contributed equally to this research.

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Table 1: Two evidence tracks.

	Track A (final data)	Track B (deployed)
Data	182 consensus clips, 20 speakers	pilot 171/17/29 + Cohort B 32
Model	calibrated multilingual intent classifier on text	Whisper-base + TF-IDF/MLP router
Evidence	12-config ablation, unseen speakers, ASR on 182	MLflow, gold eval, Docker, drift, gate
Gap	end-to-end ASR→intent not run	trained on pilot labels

2 CHANGES FROM THE PROPOSAL

Table 2: Proposal versus what was built.

Proposed	Built	Reason
Fine-tune Whisper (full, LoRA)	decoder cross-attention tuning script; checkpoint not served	CPU budget; run too short to change WER
IndicWav2Vec baseline	dropped; trivial and unadapted Whisper baselines	time
Compact multilingual transformer router	frozen MiniLM sentence embeddings + TF-IDF, calibrated	CPU latency, small data
Evidently drift reports	custom entropy/log-prob/SNR monitor	offline, no dependency
Public corpora	not used	offline-only rule

3 PART A: MODELS AND BASELINES

Track A systems. The classifier (`model/src/training/models.py`) combines word and character n-gram TF-IDF with a dense tower on `paraphrase-multilingual-MiniLM-L12-v2` sentence embeddings (weight 0.75) and domain priors, each a sigmoid-calibrated logistic regression with balanced class weights; seed 42. **Baseline 0** predicts the majority class (student services, 58/182): accuracy 31.9%, macro-F1 0.121 (computed from the class counts). **Baseline 1** is the same classifier on the raw transcripts only (C-01). Twelve configurations add augmented data (Table 3); the "proposed" full pipeline is C-10.

Track B systems. **Baseline 0**: majority intent with empty transcript (WER 1.0). **Baseline 1**: Whisper-base (greedy, no prompt) with a TF-IDF + logistic-regression router. **Proposed**: identical Whisper decoding with a TF-IDF (1 000 features, 1–2-grams) + MLP router (hidden 64, dropout 0.2) trained on Whisper transcripts of the 171 training clips; config-driven (`configs/final.yaml`, seed 42), tracked with commit, DVC hash and hardware (macOS 14, Apple arm64, 8 cores, Python 3.13).

Augmentation ablation. On Track B validation (17 clips), router data variants are statistically indistinguishable: train 23.5% [5.9, 47.1] (run `4874f30b`), train+augmented 29.4% [11.6, 52.9] (`4cb2e4e9`), train+AL-selected 23.5% (`351cc3e8`). The much larger Track-A study follows.

4 PART B: EXPERIMENT TRACKING AND TUNING

MLflow search (Track B).. Every run is logged to MLflow [8] (`supporting_files/mlflow.db`, export in `artefacts/mlflow_export/`) with parameters, validation metrics, seed, commit and artefacts. A 15-run search varies learning rate, hidden size $\{32, \ldots, 128\}$, maximum n-gram $\{1, 2, 3\}$, dropout and epochs on validation only; accuracy spans 29.4–52.9% on 17 clips (one clip = 5.9 points), so the best run 6 (`9820a2a7`; 251 KB) beats the next by a single clip (Figure 2; all runs in Appendix Table 7). Router latency is 0.005–0.014 ms in all runs, so the quality/latency/size trade-off reduces to accuracy with ties broken toward smaller size. The registry holds `production` (run 6) and `staging` (run 9, `730256f7`) aliases.

Ablation study (Track A).. The 12-configuration study (Table 3, Figure 3) is the systematic search for the final data: five folds, macro-F1, deltas versus C-01. **[TBD: log C-01..C-12 as MLflow runs and add run IDs to Table 3]**.

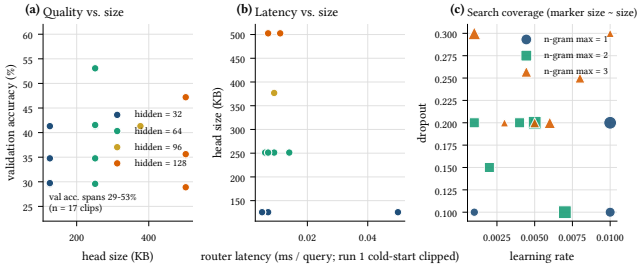


Figure 2: Search analysis over 15 MLflow runs: (a) validation accuracy vs. size, (b) latency vs. size, (c) coverage.

5 PART C: EVALUATION AND ERROR ANALYSIS

5.1 Track A: final corpus

Augmentation ablation. Table 3 and Figure 3 show macro-F1 per configuration with fold-level dots. Baseline C-01 scores 0.664 (±0.071, accuracy 69.2%). Pivot translation (C-04, +0.013), lexical synsets (C-05, +0.011) and the top-3 text combination (C-09, +0.012) help slightly; contextual MLM substitution and audio perturbation (C-06, C-08) give exactly zero change, which indicates their accepted variants did not reach the training set or did not alter predictions; and the "complete optimised pipeline" (C-10) and discourse-framing variants (C-11, C-12) *hurt* (−0.021 to −0.024). The fold standard deviation (0.06–0.08) is five times the largest gain, and paired t-tests

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across folds give $p = 0.21$–1.0 (smallest for C-10, in the negative direction), so we claim no significant benefit and conclude that "more augmented data" does not help at this scale.

Unseen speakers and language slices. On 36 clips from unseen speakers accuracy is 75.0% (Wilson CI [58.9, 86.2]) and macro-F1 0.760 (Figure 4); IT support is easiest (F1 0.92) and admissions weakest (0.67). Slices: Hindi 83.3% (12), English 75.0% (16), Gujarati 62.5% (8); no code-mixed clips are present, and the Gujarati slice interval spans 31–86%. **[TBD: regenerate independent test set (file is empty) with named unseen speakers; it is empty in the snapshot]**.

ASR on the final corpus. Whisper-base on all 182 clips gives mean WER 1.617, CER 1.266 and RTF 0.218 (Figure 4, bottom); by language Gujarati 0.55, English 1.17, code-mixed 2.08, Hindi 2.56, with fuzzy match scores 0.41, 0.60, 0.01 and 0.10. The ordering is not the expected one (Gujarati best, Hindi worst) and is the combined effect of script (Whisper outputs Latin or Urdu script for Hindi) and of reference provenance (curated native-script transcripts; Assignment 1, Section 5 of the Assignment-1 report); we therefore treat WER here as an upper-bound diagnostic. **[TBD: end-to-end ASR→intent accuracy on the 182 clips, compared with intent from reference transcripts (required by the brief)]**.

Table 3: 12-configuration ablation (5-fold macro-F1, mean ± s.d.; Δ vs. C-01; p = paired t-test over folds). Source: `model/results/ablation_study_results.json`.

ID	Training data	Macro-F1	Δ	p
C-01	Baseline (Raw Transcripts Only)	0.664 ± 0.071	+0.000	1.00
C-02	Original + Back-Translation	0.666 ± 0.065	+0.002	0.57
C-03	Original + Cross-Lingual Paraphrase	0.660 ± 0.057	-0.004	0.88
C-04	Original + Pivot Translation (Latin)	0.678 ± 0.072	+0.013	0.53
C-05	Original + Lexical Synonym Replacement	0.675 ± 0.057	+0.011	0.48
C-06	Original + Contextual MLM Substitution	0.664 ± 0.071	+0.000	1.00
C-07	Original + Campus Code-Switching	0.665 ± 0.075	+0.001	0.79
C-08	Original + Acoustic Audio Perturbations	0.664 ± 0.071	+0.000	1.00
C-09	Original + Top-3 Text Techniques	0.676 ± 0.058	+0.012	0.33
C-10	Complete Optimized Pipeline (All Gates)	0.643 ± 0.069	-0.021	0.21
C-11	Original + Indic Discourse Framing	0.640 ± 0.066	-0.024	0.27
C-12	Synergistic Ensemble (Latin Pivot + Indic Discourse)	0.647 ± 0.075	-0.017	0.26

5.2 Track B: deployed pipeline on pilot gold

Protocol. `evaluate.py--modelproduction--split<split>` reports WER, CER, intent accuracy and macro-F1 with 1 000-resample bootstrap intervals [1] (seed 42). Paired tests: Wilcoxon [7] on per-clip WER and McNemar [4] on intent. Test and gold are the same 29 clips.

Findings. (i) WER exceeds 1 on every split. On gold, substitutions are 0.83, deletions 0.16 and insertions 0.30 per reference word and only 14 of 838 reference words (1.7%) are matched (Figure 5); an empty output scores 1.0, so this ASR stage is worse than doing nothing. (ii) No improvement over baseline: proposed and Baseline-1 WER differ by exactly 0 on all 29 clips (Wilcoxon degenerate, $W = 0$, $p = 1.0$) and McNemar on intent gives $\chi^2 = 0.10$, $p = 0.75$. (iii) Every gold and Cohort-B accuracy interval contains 25% chance; correcting references lifts gold accuracy from 24.0% to 30.8% with WER still 1.3, so the ASR output is the bottleneck.

Slices and errors. On gold, `campus_outdoor` (6 clips, 16.7%) and Gujarati–English (13 clips, 23.1%) are weakest and Rajasthan (6 clips, 66.7%) strongest; all slices have $n \leq 16$ so Wilson intervals span ≥ 40 points (Appendix Table 8). We categorise 20 errors with fixes (Appendix Table 9; Figure 6): the dominant mode is a fluent hypothesis about another topic, then script collapse (Devanagari/Urdu tokens), near-silent or truncated output, phonetic substitution of Gujarati words and genuine cross-intent overlap. Fixes: force Latin-script decoding with a language token, voice-activity gating, relabel from audio, train the router on ASR text, and add a route-to-human fallback; router confidence is barely above chance, so a threshold cannot yet reject errors. `MODEL_CARD.md` documents use, data, intervals, slices, limits and risks.

6 PART D: DEPLOYMENT AND INTERFACE

Service. FastAPI exposes `POST /predict`, `GET /health` and `GET /version` (commit and model version); the Streamlit UI offers upload/microphone input, bundled consented examples, confidence, version, latency, a monitoring panel and a validation panel comparing reported and measured values. Requests are logged without audio or personal data (`supporting_files/api_access.log`); eight pytest tests cover normalisation, the gold hash, gate logic and the endpoints.

Image and CI. `code/Dockerfile` builds from a digest-pinned `python:3.10-slim`, upgrades `pip/setuptools<70`, installs pinned requirements, caches Whisper-base weights at build time, bundles code, data, artefacts and the gold set, and records the commit as a build argument. Start: `docker run –rm -p 8501:8501 -p 8000:8000 <image>`; validate: `docker run –rm –network none <image> validate`. CI (`.github/workflows/ci.yml`) runs tests, sample validation, an image build and an offline smoke request.

Performance and offline evidence. Measured on Apple arm64 over the 29 gold clips (523.8 s audio): p50 2.86 s, p90 7.69 s, p95 13.44 s, mean 4.39 s; RTF 0.243; throughput 0.23/s; peak memory 783 MB (Figure 7). `TEST_SET_LOG.md` (LOG-016) records an offline validation with matching gold checksum (WER 1.358, CER 0.918, accuracy 24.1%, macro-F1 0.221), but its JSON lists a macOS host and `offline_network_isolated: false`, so it is a host-side validation, not proof of a `–network none` container run. Tolerance for the validator against Table 4: ±0.10 WER, ±1 clip. **Still to record from an actual `linux/amd64` container:** image size **[TBD: image size GB]**, p50/p95 **[TBD: container p50/p95 ms]**, peak RAM **[TBD: container peak RAM]**, archive SHA-256 **[TBD: docker_image.tar.gz SHA-256]** and the verbatim output of the `–network none validate` run.

7 PART E: MONITORING, DRIFT AND RETRAINING

Replay. `traffic_replay.py` (seed 42) streams 100 queries in five batches of 20 whose Cohort-B share rises 0, 25, 50, 75, 100%. Two label-free signals (router entropy, decoder average log-prob),

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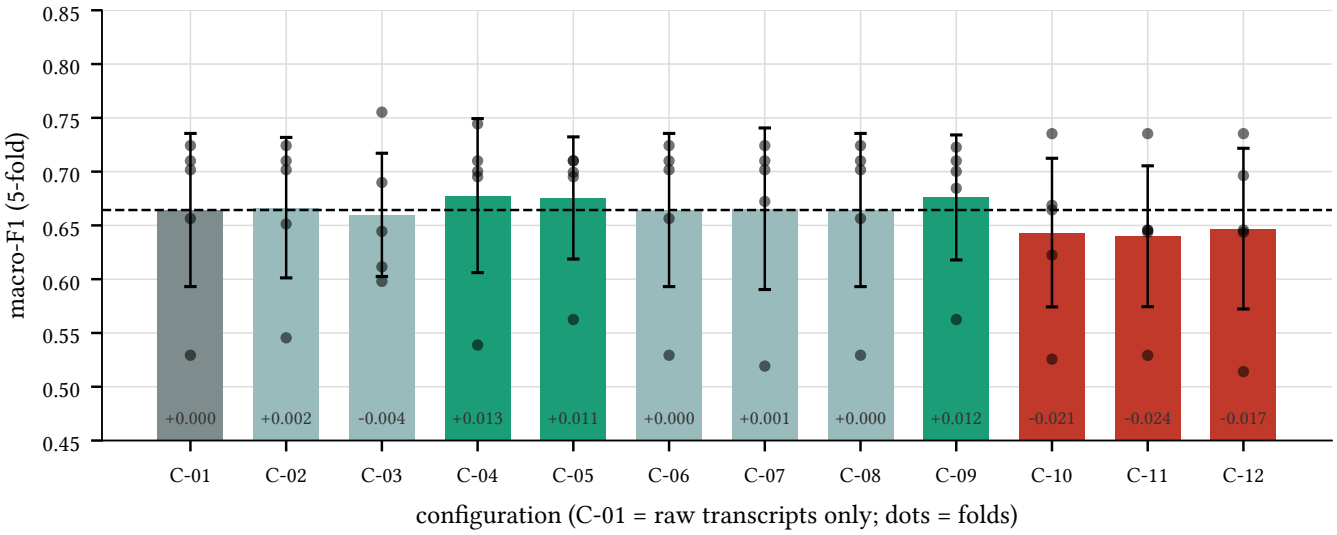


Figure 3: Ablation over 12 data configurations (bars: mean ± s.d.; dots: folds; numbers: Δ vs. C-01). Green = gain > 0.005, red = loss < −0.005.

Table 4: Track B: proposed system with 95% bootstrap CIs (B1 = Baseline 1 intent accuracy; WER/CER equal for proposed and B1). Source: `supporting_files/evaluation_results.json`.

Split (*n*)	WER	CER	Intent acc. (%)	Macro-F1	B1 acc. (%)
Validation (17)	1.403 [1.225, 1.601]	1.050 [0.968, 1.143]	53.5% [29.4, 76.5]	0.451 [0.226, 0.637]	29.7% [11.8, 52.9]
Gold, original (29)	1.272 [1.156, 1.393]	0.945 [0.891, 1.006]	24.0% [10.3, 41.4]	0.225 [0.083, 0.389]	31.0% [13.8, 48.3]
Gold, corrected (29)	1.298 [1.088, 1.501]	1.008 [0.855, 1.150]	30.8% [13.8, 48.3]	0.261 [0.127, 0.393]	41.5% [24.1, 58.6]
Cohort B (32)	1.104 [1.040, 1.187]	0.947 [0.900, 0.994]	25.2% [12.5, 40.6]	0.219 [0.089, 0.359]	25.2% [12.5, 40.6]
Augmented (60)[†]	1.684 [1.590, 1.786]	1.060 [1.012, 1.113]	63.5% [51.7, 75.0]	0.613 [0.488, 0.738]	26.6% [16.7, 38.3]

[†]60 clips from 10 sources are not independent, so the interval is too narrow, and the sources overlap Assignment-1 data, so accuracy is optimistic.

SNR, and one labelled metric (intent accuracy) are tracked; the entropy threshold is the 95th percentile of validation entropy (1.662 bits).

Result: no usable detection. Batch entropy never exceeds the threshold (1.21–1.35), accuracy does not fall with drift (35, 35, 15, 25, 45%) and correlations with accuracy over five batches are 0.41 (entropy), 0.20 (log-prob) and 0.20 (SNR)—none significant (Figure 8). The log reports an alert at batch 5, but the batch-alert expression contains an unconditional clause for batch ≥ 5, so that alert reflects the schedule, not a signal; we do not count it as detection, and the logged false-alarm rate (0/17) is a constant in the script. A valid design needs many seeded in-domain replays, a sequential test such as CUSUM or Page–Hinkley [2], and shift-sensitive signals (non-Latin-script fraction, hypothesis-length ratio) [5].

Retraining pipeline and gate. One command (`retrain_pipeline.py --mode rehearsal|no_rehearsal`) selects the 12 highest-entropy clips from the P4 adaptation pool, reveals their labels (simulated annotator), validates, retrains (rehearsal: 171+12; no-rehearsal: 12 only, a forgetting control [3, 6]) and evaluates on held-out P13 (13 clips) and gold. `configs/gate.yaml`: promote only if Cohort-B accuracy rises by ≥ 5 points and gold falls by ≤ 4 points; every gold evaluation is appended to the test log.

Outcomes. The final run (LOG-179/188) *promotes* the rehearsal candidate (Cohort B 30.8%→38.5%, 4→5 of 13; gold 24.1%→20.7%, 7→6 of 29) and *rejects* the no-rehearsal candidate (no gain), with rollback (Figure 9). The decision hinges on one clip each side; an earlier run at the same commit (LOG-418) rejected the same candidate, and the gold tolerance was widened from 2 to 4 points during development. The gate therefore demonstrates the mechanism—promote, reject, roll back, log—and is not evidence that retraining helps; it should be frozen in a committed file before evaluation and judged with a paired interval.

8 PART F: REPRODUCIBILITY

Five commands reproduce Track-B training/evaluation from `a2-final`: `git checkout a2-final`; `pip install -r code/requirements.txt`; `dvc pull`; `python code/train.py --config configs/final.yaml`; `python code/a2_part_c_evaluation/evaluate.py --split gold`. Track A: `python -m model.cli ablate` and `benchmark`. Docker commands, versions, seeds (42), checksums and registry versions

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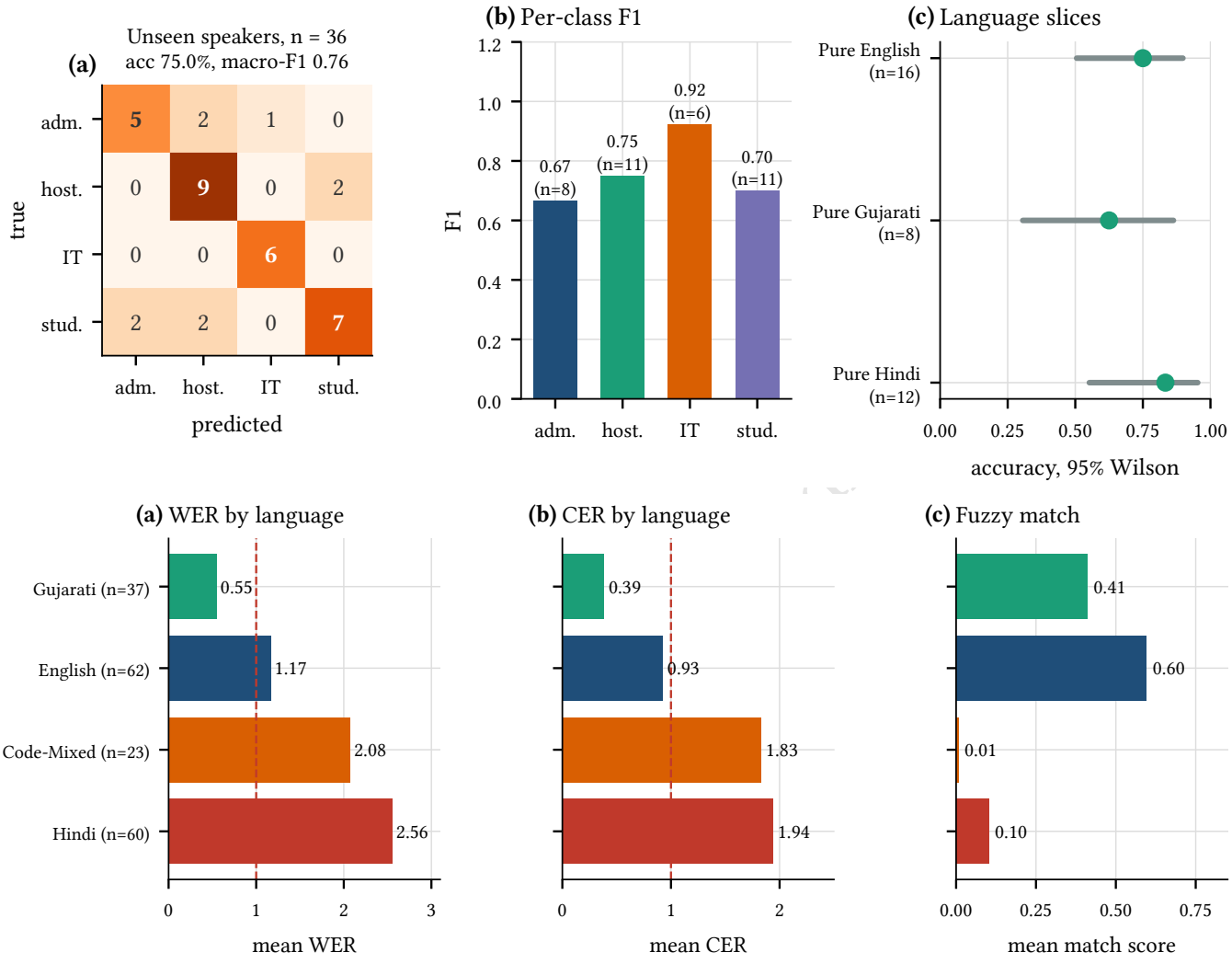


Figure 4: Top: unseen-speaker benchmark (36 clips): confusion matrix, per-class F1, language slices with Wilson intervals. Bottom: Whisper-base on 182 clips by language (WER, CER, fuzzy match).

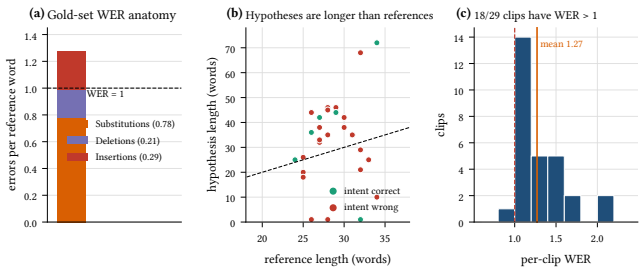


Figure 5: Gold WER anatomy: (a) substitutions/deletions/insertions per reference word, (b) hypothesis vs. reference length, (c) per-clip WER (19/29 above 1).

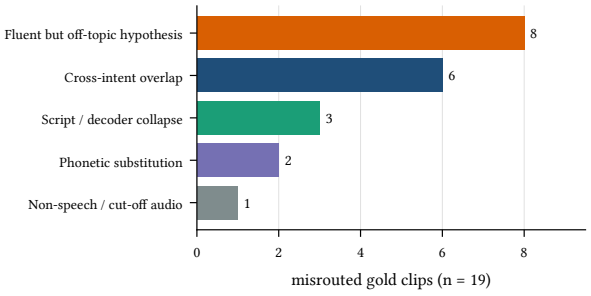


Figure 6: Error taxonomy (20 sheet rows, regrouped).

`report_numbers.py` regenerates documentation numbers. Track A's are in `DOCKER.md`, `ENVIRONMENT.md` and `MODELS.md`; `scripts/generate_sentence-embedding` model is fetched by `sentence-transformers`;

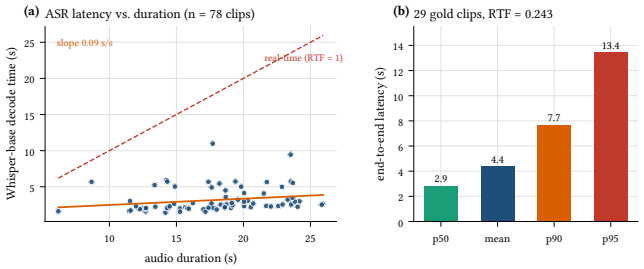


Figure 7: (a) Whisper-base decode time vs. audio duration (78 clips). (b) End-to-end latency percentiles on 29 gold clips.

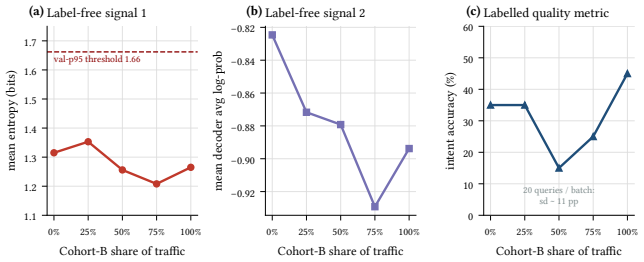


Figure 8: Replay under shift: (a) entropy with the validation threshold, (b) decoder log-prob, (c) accuracy per batch of 20.

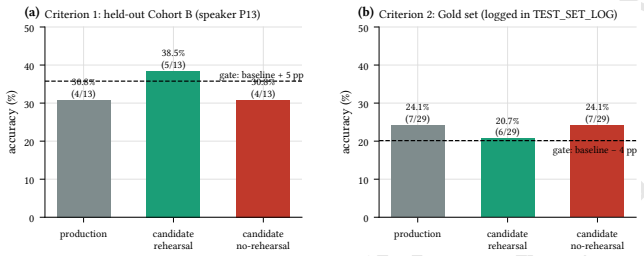


Figure 9: Promotion gate: (a) held-out Cohort-B accuracy, (b) gold accuracy for production, rehearsal and no-rehearsal candidates.

[TBD: bundle the weights in the image so that no runtime download occurs].

9 LIMITATIONS AND CONCLUSION

Limitations. (1) The deployed ASR is unadapted Whisper-base; the tuning run (5 steps) is too short to matter and its checkpoint is not served. (2) Track A models read curated reference text, not ASR output, and are not yet integrated into the container. (3) Reference provenance: transcripts come from a curated mapping (Assignment 1), so WER is a diagnostic. (4) Test sets are tiny (13–36 clips), so intervals are wide. (5) Folds are clip-grouped, not speaker-grouped. (6) The replay alert is schedule-driven and false alarms are unmeasured. (7) Container measurements are pending. (8) Gate history weakens the retraining claim. **Next steps.** Fine-tune the Whisper decoder for hundreds of steps and serve it; train the router on ASR text of the final corpus and report end-to-end intent accuracy; rebuild speaker-disjoint folds; replace the drift rule with a validation-calibrated sequential test and a measured false-alarm rate; freeze the gate before evaluation; record container measurements. **Conclusion.** The tracking, evaluation, serving, monitoring and gating infrastructure is complete and auditable, and the evidence is reported honestly: augmentation does not significantly help at this scale, the intent classifier generalises to unseen speakers at 75%, and the deployed ASR does not yet beat trivial baselines.

Table 5: Contribution summary (equal 33.3% each).

Member	Lead areas (Assignment 2)
R. Vraj	**[TBD: fill in]**
V. S. Patil	**[TBD: fill in]**
S. Sharma	**[TBD: fill in]**

Member contributions.

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A EVIDENCE INDEX

Table 6: Evidence index (paths from repository root).

Result	Supporting file / run
Ablation (Table 3), p-values	`model/results/ablation_study_results.json` (fold F1s)
Unseen speakers	`model/results/unseen_speaker_benchmark.json`
ASR on 182 clips	`model/data/reports/asr_evaluation_report.json`
Track-B metrics (Table 4)	`supporting_files/evaluation_results.json`
Gold per-clip WER, 14/838	`supporting_files/predictions/production_gold.csv`
Data ablation (B)	`supporting_files/part_a_baselines_benchmark.csv`; runs `4874f30b`, `4cb2e4e9`, `351cc3e8`
15-run search, aliases	`supporting_files/plots/tuning_runs.json`; `9820a2a7` (prod.), `730256f7` (stag.)
20 errors	`supporting_files/error_analysis.csv`
Latency, RTF, memory	`supporting_files/loadtest/latency_profile.json`
Offline validation	`supporting_files/offline_validation_output.json`; `TEST_SET_LOG.md` LOG-016
Replay, correlations	`supporting_files/traffic_replay_log.json`
Gate outcomes	`TEST_SET_LOG.md` (LOG-179, 188, 418); `configs/gate.yaml`
Gold corrections	`data/gold_corrections.csv`
Figure code	`code/scripts/make_report_figures.py`, `make_report_figures_final.py`

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Table 7: All 15 MLflow search runs (validation only); full IDs in `supporting_files/plots/tuning_runs.json`.

Run	lr	Hidden	n-gram	Dropout	Epochs	Val acc. (%)	Macro-F1	Size (KB)	MLflow run
1	0.001	32	1	0.1	25	29.4	0.200	126	`61d8d420`
2	0.005	32	2	0.2	25	29.4	0.236	126	`d01a4cb5`
3	0.01	32	3	0.3	25	41.2	0.371	126	`b4e145fd`
4	0.001	64	2	0.2	30	41.2	0.343	251	`fd0ac5ab`
5	0.005	64	3	0.2	30	41.2	0.368	251	`f47bfe22`
6	0.01	64	1	0.1	30	52.9	0.470	251	`9820a2a7` (prod.)
7	0.001	128	3	0.3	35	35.3	0.309	503	`75581fbf`
8	0.005	128	2	0.2	35	35.3	0.293	503	`4f2e2cd1`
9	0.01	128	1	0.2	35	47.1	0.379	503	`730256f7` (stag.)
10	0.002	64	2	0.15	30	35.3	0.309	251	`9e00b3d8`
11	0.008	64	3	0.25	30	41.2	0.383	251	`03e70444`
12	0.003	32	3	0.2	25	35.3	0.303	126	`64cd9b69`
13	0.007	128	2	0.1	35	29.4	0.267	503	`984be4ec`
14	0.004	64	2	0.2	28	29.4	0.236	251	`703dc1b4`
15	0.006	96	3	0.2	32	41.2	0.368	377	`3130621f`

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INSERT SCREENSHOT:
shots/ui_prediction.png
Streamlit UI: prediction with confidence, version, latency

INSERT SCREENSHOT:
shots/mlflow_runs.png
MLflow UI: 15-run comparison and registry aliases

INSERT SCREENSHOT:
shots/docker_validate.png
terminal: docker run –network none ... validate

INSERT SCREENSHOT:
shots/ci_green.png
GitHub Actions: all steps green

Figure 10: Evidence screenshots (files under `report/shots/`). The 3-minute recording goes in `supporting_files/`.

Table 8: Gold-set slices (original labels).

Dimension	Slice	n	Acc. (%)	WER
Noise	campus_outdoor	6	16.7	1.64
Noise	corridor_reverb	13	23.1	1.17
Noise	quiet_room	10	30.0	1.18
Language	gu-en	13	15.4	1.36
Language	hi-en	16	31.2	1.20
Region	Gujarat	13	15.4	1.36
Region	North India	10	30.0	1.18
Region	Rajasthan	6	33.3	1.22
Device	OnePlus Nord 2	6	16.7	1.64
Device	Realme Narzo 30	7	14.3	1.12
Device	Samsung Galaxy M52	10	30.0	1.18
Device	Xiaomi Redmi Note 11	6	33.3	1.22

B USE OF AI ASSISTANCE

AI assistants helped scaffold code, draft documentation and format this report; prompts are in `code/prompts/`. Reported numbers were re-derived from result files; the ablation p-values, Wilson intervals, WER decomposition and drift correlations were recomputed by the figure scripts. **[TBD: exact assistant names/versions]**

C SCREENSHOTS AND RECORDINGS TO INSERT

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Table 9: Error sheet (20 errors, truncated; full text in `error_analysis.csv`).

ID	True→pred.	WER	Category	Suggested fix
ERR-01	hostel_mess→it_support	1.19	Phonetic & Lexical Substitution	Add Hinglish/Gujlish vocabulary expansion in tokenizer and train intent head on character-n-gra…
ERR-02	it_support→hostel_mess	1.00	ASR Decoder Hallucination (Script Shift)	Set condition_on_previous_text=False and compression_ratio_threshold=2.0 in Whisper decoding op…
ERR-03	student_services→hostel_mess	1.00	Acoustic Silence / Non-Speech Collapse	Prepend energy-based voice activity detection (VAD) filter to reject low-energy frames before W…
ERR-04	admissions→hostel_mess	1.00	ASR Decoder Hallucination (Script Shift)	Force Whisper language='en' decoding task with Latin script constraint to prevent unprompted Hi…
ERR-05	student_services→it_support	1.64	Lexical Cross-Talk & Hallucination	Use contextual embeddings (e.g. MiniLM / IndicBERT) that score entire query semantics rather th…
ERR-06	student_services→it_support	1.58	Semantic Hallucination Overwrite	Fine-tune Whisper acoustic decoder on target speaker acoustics to prevent severe phantom halluc…
ERR-07	admissions→hostel_mess	2.12	Semantic Hallucination Overwrite	Incorporate audio-level intent classification directly from Whisper encoder embeddings (end-to-…
ERR-08	hostel_mess→it_support	1.61	Semantic Keyword Hijack	Apply cross-attention acoustic adaptation on speaker cohort to ensure domain-specific hostel vo…
ERR-09	it_support→student_services	1.00	Cross-Domain Semantic Misalignment	Add domain-specific lexicon biasing during Whisper beam search decoding using campus vocabulary…
ERR-10	admissions→it_support	2.12	Lexical Confusion & Peripheral Tokens	Train joint ASR-Intent loss where acoustic encoder explicitly weights intent discriminative pho…
ERR-11	it_support→admissions	1.00	Cross-Intent Academic Overlap	Disambiguate academic deadline queries from administrative admissions via fine-grained multi-la…
ERR-12	hostel_mess→it_support	1.22	Acoustic Bleed & Academic Hallucination	Normalize Whisper prompt conditioning to bias toward residential and campus administration term…
ERR-13	it_support→student_services	1.00	Cross-Intent Academic Overlap	Weight domain-specific technical IT terms ('email forwarding', 'hpc', 'server') higher using we…
ERR-14	student_services→it_support	1.38	Acoustic Substitution & Technical Drift	Expand medical/counselling training samples in student_services intent class to increase semant…
ERR-15	admissions→hostel_mess	1.40	Severe Semantic Hallucination (Hostel Bias)	Fine-tune Whisper base decoder cross-attention layers on domain recordings to suppress random r…
ERR-16	hostel_mess→student_services	1.25	Composite Intent Ambiguity	Introduce composite intent schema or fallback routing to human agent when classification entrop…
ERR-17	it_support→admissions	0.97	Hallucinated Guest Accommodation Query	Fine-tune acoustic encoder on noisy lab acoustic environments with server room background rumbl…
ERR-18	student_services→hostel_mess	1.56	Academic Schedule Confusion	Calibrate probability thresholds per intent class to avoid majority class collapse on uncertain…
ERR-19	admissions→student_services	1.21	Multilingual Script Infiltration (Urdu/Hinglish)	Enforce strict Unicode Latin/Devanagari script filtering in ASR post-processing pipeline before…
ERR-20	admissions→student_services	1.06	Acoustic Mismatch & Keyword Ambiguity	Implement multi-task loss weighting and negative sample penalty on cross-domain vocabulary.…

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